

AI5707: Pattern Recognition and Machine Learning

Assignment 2: Image Reconstruction using EVD and SVD

1. Introduction

The aim of the task was to study low-rank matrix approximation for image reconstruction and compression.

2. Experiments Performed

2.1 Task 1: EVD and SVD on a square image

The square image, with dimensions of 512×512 pixels was converted from RGB to grayscale to obtain a matrix A . EVD was computed using `numpy.linalg.eig(A)` function, and SVD was computed using `numpy.linalg.svd(A)` function. Both eigenvalues and singular values were sorted in decreasing order of magnitude before truncation, hence the top- k values meant the k most significant components.

A grayscale image matrix is usually not symmetric, hence its eigenvalues and eigenvectors from EVD can be complex. Rather than working with complex numbers by removing imaginary parts once the computations were completed, the reconstruction was built using pure real arithmetic throughout, via the standard real-canonical-form identity; since A is real, any complex eigenvalue occurs as a conjugate pair $a \pm bi$ with conjugate eigenvectors $p \pm iq$ (where p and q are real vectors), and comparing the real and imaginary parts of $A(p+iq) = (a+ib)(p+iq)$ gives $Ap = ap - bq$ and $Aq = bp + aq$. This means that, in the real basis $\{p, q\}$, A acts exactly like the real 2×2 matrix $\begin{bmatrix} a & b \\ -b & a \end{bmatrix}$. Using this, a fully real basis matrix Q_{real} and a real block-diagonal Λ_{real} were built directly, with each real eigenvalue contributing one real column and a 1×1 block; each conjugate pair contributing two real columns, for the vectors p and q , and a 2×2 block. The full reconstruction is of float64 datatype throughout.

Because both members of a conjugate pair have the same magnitude, they are adjacent after sorting by magnitude, hence each pair is treated as one indivisible block when choosing the top- k . If the requested k falls in the middle of a pair, it is rounded up to the end of that block. In this image, 474 of the 512 eigenvalues are complex, as a result out of the 3 k values used, the $k = 50$ was adjusted to $k = 51$, and $k = 100$ to $k = 101$. The final k value, $k = 160$, did not need any such adjustments.

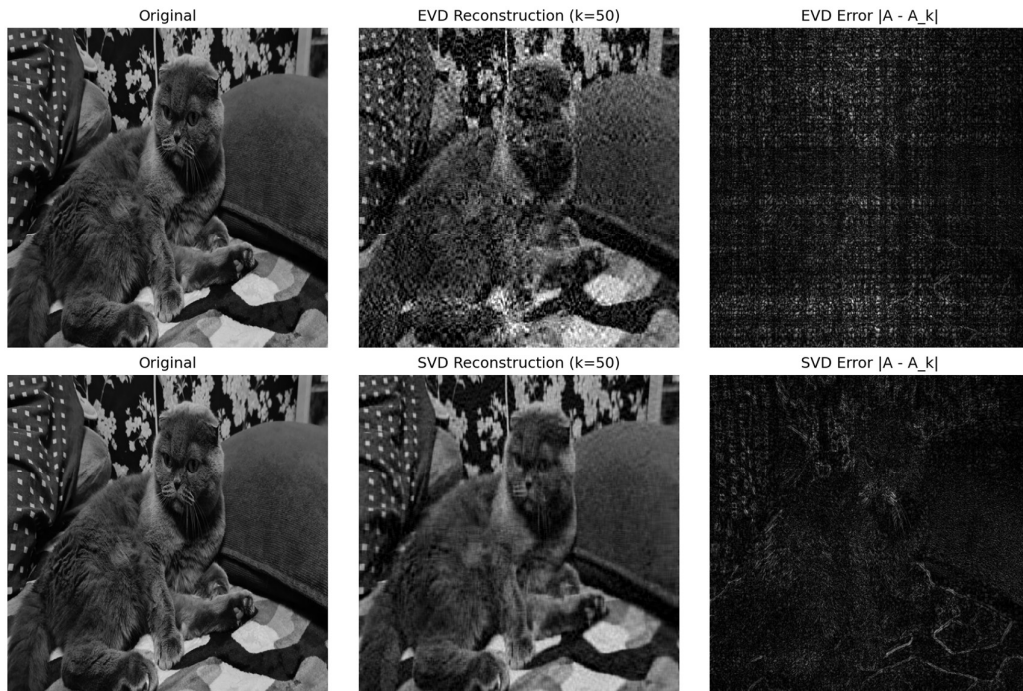


Figure 1: Original, EVD and SVD reconstructions, and error maps for $k = 50$.

At $k = 50$, both reconstructions are quite degraded, but the SVD reconstruction can already be seen capturing the coarse black and white layout of the cat and background, while the EVD reconstruction is dominated by noisy, high-contrast artefacts with much less recognisable structure. The error map for EVD is bright, signifying large amounts of error present everywhere, whereas the SVD error map is darker when compared to EVD.

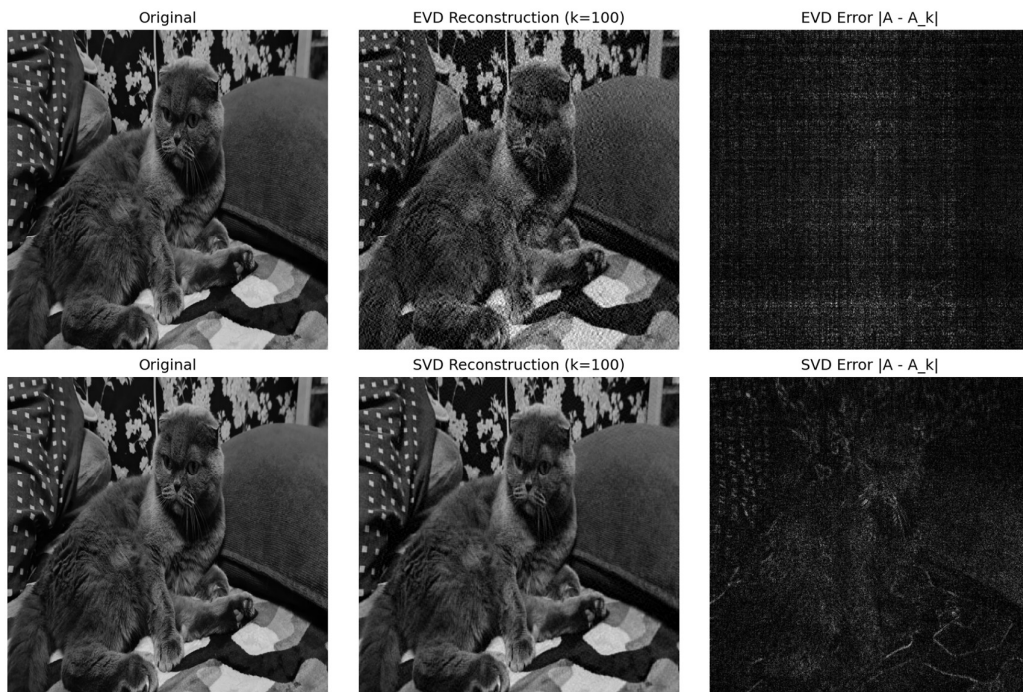


Figure 2: Original, EVD and SVD reconstructions, and error maps for $k = 100$.

At $k = 100$, the SVD reconstruction is visually close to the original: the cat's outline, face and the patterned background are clearly recognisable, with only the finer texture missing. The EVD reconstruction has improved and the cat is now recognisable, but it still shows noticeable grid-like noise superimposed on the image, and its error map remains brighter than SVD's.

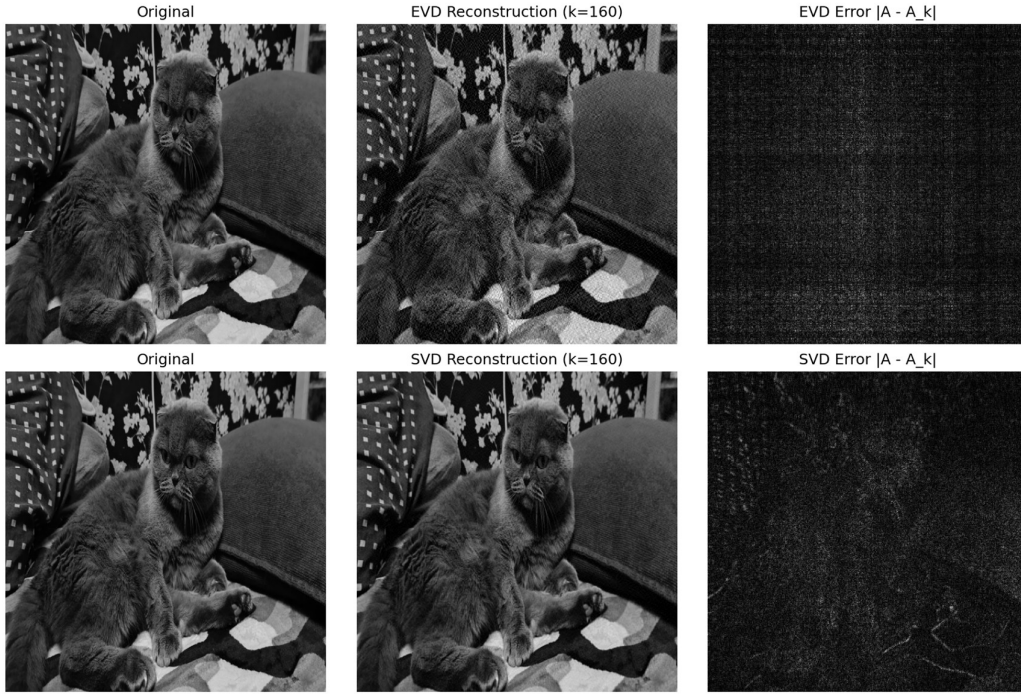


Figure 3: Original, EVD and SVD reconstructions, and error maps for $k = 160$.

At $k = 160$, the SVD reconstruction is nearly indistinguishable from the original, with error concentrated only along the finer edges and texture. The EVD reconstruction has also improved considerably and now resembles the cat clearly, but the residual noise pattern and a visibly brighter error map show that, for the same k , EVD still reconstructs less accurately than SVD.

The Frobenius norm errors $E(k)$ computed for these three values of k were (EVD's k adjusted where needed to keep conjugate pairs together, as explained above):

k	EVD error $E(k)$	SVD error $E(k)$
51	13370.34	5532.17
101	9514.12	3323.29
160	7745.87	2063.52

We observe that EVD's error does not decrease monotonically as k increases. This is because, unlike SVD, EVD's components are not built from an orthonormal basis, so adding one more term to the sum is not guaranteed to reduce the Frobenius error. This is the same reason the EVD error curve in Figure 4 is noisy and non-monotonic rather than smoothly decreasing.

Finally, $E(k)$ was computed for every possible value of k from 1 to 512 and plotted for both methods on the same axes (Figure 4).

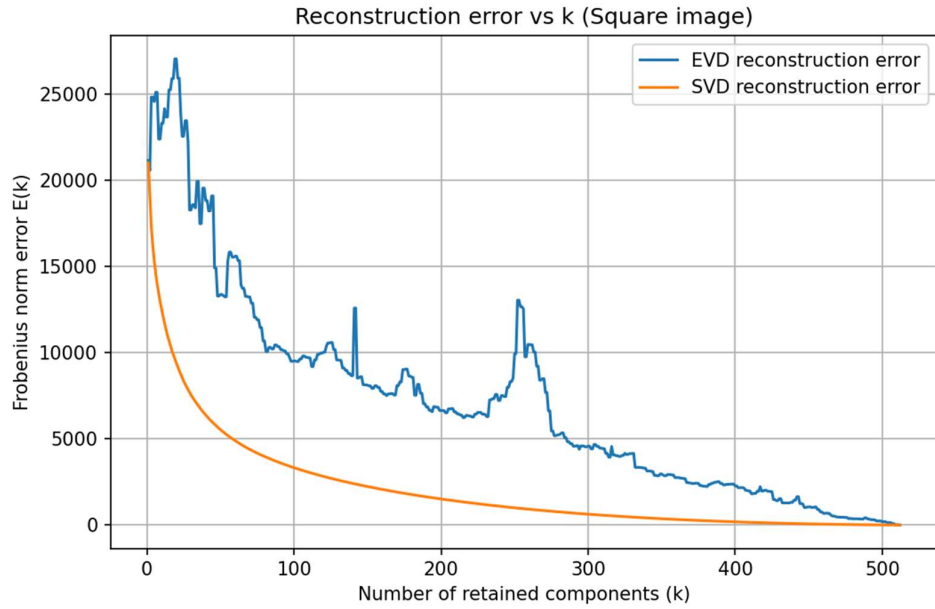


Figure 4: Frobenius norm reconstruction error $E(k)$ versus k for EVD and SVD (square image).

The SVD curve decreases smoothly and monotonically to zero as k approaches 512, which is expected because singular values are non-negative and sorted in decreasing order, so each additional component either reduce or at worst keep the reconstruction error unchanged. The EVD curve, in contrast, decreases only on average; it is noisy and non-monotonic, with visible spikes where the error temporarily increases as k grows. Both curves reach zero error at $k = n$, since retaining all components reconstructs A exactly.

2.2 Task 2: SVD on a rectangular image

The group's rectangular image, with dimensions 301×360 pixels was converted to grayscale to obtain the matrix A . Since A is not square, it does not have an eigenvalue decomposition, so only SVD ($A = U\Sigma V^T$, computed with `numpy.linalg.svd`) was applied, giving 301 singular values, since 301 is the lower of 301 and 360. The same three values of $k = 50, 100, 160$ were used as with Task 1.



Figure 5: Original, SVD reconstruction, and error map for $k = 50$ (rectangular image).



Figure 6: Original, SVD reconstruction, and error map for $k = 100$ (rectangular image).



Figure 7: Original, SVD reconstruction, and error map for $k = 160$ (rectangular image).

As with the square image, the reconstruction quality improves steadily with k : at $k = 50$ the coarse layout and the cat are already recognisable with visible texture loss, at $k = 100$ most of the finer texture is recovered, with errors remaining mainly along edges, and at $k = 160$ (roughly half of the 301 available singular values) the reconstruction is visually almost identical to the original, with error confined mainly to the finer edges (whiskers, fur texture, chair pattern).

The Frobenius norm errors obtained were as follows: At $k = 50$, 1853.81, $k = 100$, 878.57, $k = 160$, 355.02.

$E(k)$ was then computed for every k from 1 to 301 and plotted (Figure 8).

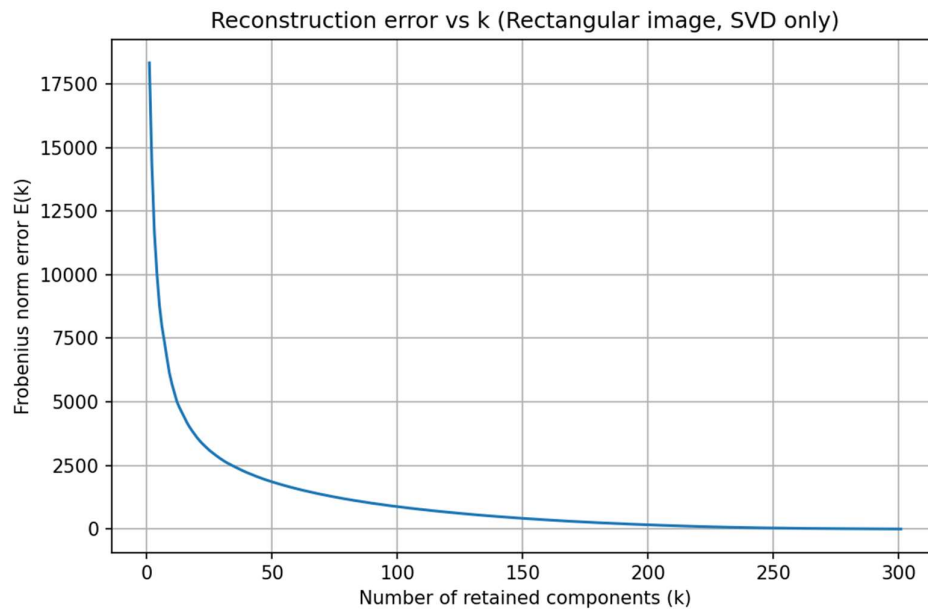


Figure 8: Frobenius norm reconstruction error $E(k)$ versus k for SVD (rectangular image).

This curve is smooth and monotonically decreasing, just like the SVD curve in Task 1, reaching zero once k approaches 301, the full rank of the matrix.

3. Inference

- SVD gives better reconstruction than EVD for the same k .
- SVD's error curve $E(k)$ decreases smoothly and monotonically.
- EVD's error curve is noisy and sometimes increases locally as k grows. This is because the matrix of eigenvectors, Q is generally not orthogonal for a non-symmetric image matrix A , so $Q^{-1} \neq Q^T$.
- For both decompositions, lower values of k retain only the coarse, low-frequency structure of the image, with higher values of k values progressively adding back finer detail, such as edge and texture; at $k = n$, the reconstruction exactly equals the original image and $E(k) = 0$.
- Because SVD reaches low error at much smaller k than EVD, SVD is the better practical choice for image compression: a small number of singular values/vectors can represent most of the visual information, which is not true of eigenvalues/eigenvectors for a general (non-symmetric) image matrix.
- EVD is only directly applicable to square matrices, so for the rectangular image only SVD could be used